

4.6 Solving Problems Involving Rectangles on the Coordinate Plane

Lesson Introduction

 Benchmarks	MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane. MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle.		 Learning Targets	- I can find distances between ordered pairs. - I can find the area of a rectangle on a coordinate plane. - I can find the fourth vertex of a rectangle on a coordinate plane.
 Benchmark Coherence	Building On MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.		Working Toward MA.912.GR.3.4 Use coordinate geometry to solve mathematical and real-world problems on the coordinate plane involving perimeter or area of polygons.	
 Essential Questions	- How can you find the area of a rectangle on the coordinate plane? - How can you find the missing vertex of a rectangle on the coordinate plane?			
 Lesson Narrative	In this lesson, students will use their knowledge of the coordinate plane to help them find the area of rectangles. They will also use this knowledge to find the missing vertex of a rectangle on the coordinate plane. Students should use their prior knowledge from the previous lessons on distance on the coordinate plane, as well as what we know about the properties of a rectangle, throughout this lesson.			
 Component Summary	4.6.1 Warm-Up (~5 min.)	Complete a multiplication equation to make it true (<i>SE p. 166</i>)	4.6.4 Collaboration (10 min.)	Discuss the attributes of a rectangle and how they relate to the coordinate plane (<i>SE p. 168</i>)
	4.6.2 Collaboration (5-10 min.)	Find the area of a rectangle on the coordinate plane (<i>SE p. 166</i>)	4.6.5 Guided Practice (5-10 min.)	Determine the missing vertex of a rectangle (<i>SE p. 169</i>)
	4.6.3 Guided Practice (5-10 min.)	Find the area of rectangles on the coordinate plane (<i>SE p. 167</i>)	4.6.6 Your Turn! (5-10 min.)	Determine the missing vertex and area of a rectangle (<i>SE p. 169</i>)
	4.6.7 Wrap-Up (~5 min.)	Reflection on potential learning gaps (<i>SE p. 170</i>)		
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Vertex - Parallel - Area	(Optional) Colored pencils (Optional) Graph paper	N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students might confuse x- and y-values in the coordinate plane. - Students may confuse when to add or when to subtract when finding distance on the coordinate plane.	- Students may wonder which side to call length and which side to call width. If these are not noted, then it should not matter which is named. Sometimes a general rule, such as length is the longer side and width is the shorter side is used in a classroom, but if this is done, it should be noted as a matter of convenience and not as a rule. However, the goal of the lesson is for students to find the lengths of the sides and use them to find area. At this point, it is not important which side they refer to as width and which they refer to as length. - Consider providing sentence frames for students to explain their reasoning.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder In 4.6.4 Collaboration, students engage in a discussion about the attributes of a rectangle on the coordinate plane. This discussion could be used as a Notice and Wonder to position students to think about the x -coordinates of the points that create a vertical line and the y -coordinates of the points that make a horizontal line and how they relate.	MA.K12.MTR.5.1: Patterns and Structure Students should relate what they already know about rectangles, i.e., what makes a shape a rectangle, finding area, etc., and apply that knowledge to finding the missing vertex and area of a rectangle on the coordinate plane.
 Differentiate Instruction	This lesson provides multiple visual entry points through graphs on the coordinate plane. Students use the grid to find missing vertices in the rectangles and can also use the visual of the rectangle to reason about the area. Auditory entry points are provided through collaborative activities and partner discussion. Consider how best to pair students for each Collaboration (4.6.2 and 4.6.4) to ensure discussions are fruitful and impactful to understanding. Data collected in the previous lesson can be used to inform partner pairs or small groups.	
Lesson Notes:		

4.6.1 Warm-Up: True Statement

Component Overview: Students fill in the boxes to make the multiplication equation true, using digits one through nine.



4.6.1 Warm-Up: True Statement

1. Place a digit in each blue box to make the statement true. Use only digits 1 to 9, at most one time each.

$$\boxed{1}\boxed{5} \times 3.2 = \boxed{4}\boxed{8}$$



Consider using this Warm-Up as an opportunity for informal observation of student proficiency in multiplying decimals. Students will be finding area using rational numbers throughout this lesson, and informal data from this Warm-Up can inform partner pairs or small groups for the 4.6.2 and 4.6.4 Collaboration activities later on.

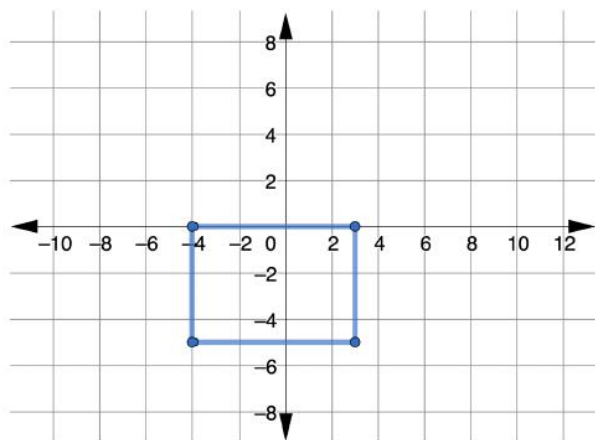
4.6.2 Collaboration: Area of Rectangles

Component Overview: Students collaborate with a partner to review the formula for area of a rectangle. Students then use the formula to find the area of a rectangle on the coordinate grid.



4.6.2 Collaboration: Area of Rectangles

1. Work with your partner to explain how to find the area of a rectangle.
Area of a rectangle = Length $\times$ Width
2. A rectangle with coordinate $(3, 0)$, $(3, -5)$, $(-4, 0)$, and $(-4, -5)$ is shown. The scale of the grid is in centimeters (cm).



Work with your partner to answer the following questions.

- a. Determine the length and the width of the rectangle. Include units.
Length = 5 cm
Width = 7 cm
- b. What is the area of the rectangle? Include units.
 $A = 7 \times 5 = 35 \text{ cm}^2$



Consider highlighting student responses that represent finding area as an equation, to encourage mathematical fluency as students answer questions with area throughout this lesson.

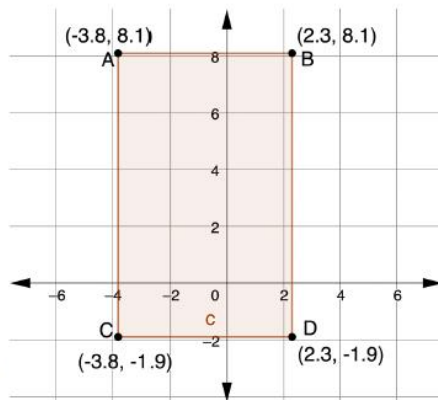
4.6.3 Guided Practice: Finding the Area of Rectangles on the Coordinate Plane

Component Overview: Students find the areas of rectangles on the coordinate plane. Encourage students to apply what they have learned in the last few lessons on finding distance between points on the coordinate plane, as well as finding perimeters of rectangles on the coordinate plane.



4.6.3 Guided Practice: Finding the Area of Rectangles in the Coordinate Plane

- Henry wants to install new carpet in his living room. A drawing of his living room, with measurements in feet, is shown.



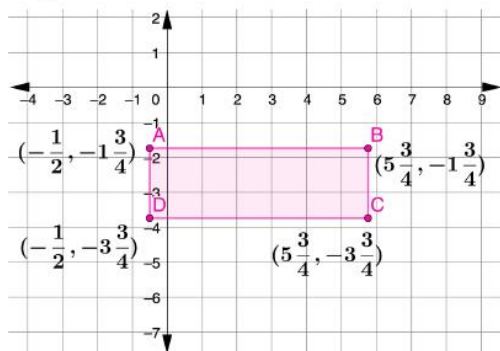
What is the area of the living room?

Length of room = 10 ft

Width of room = 6.1 ft

Area = $6.1 \times 10 = 61 \text{ ft}^2$

- A rectangle is shown.



What is the area of the rectangle?

Length of rectangle = 2 units

Width of rectangle = $6\frac{1}{4}$ units

Area of rectangle = $6\frac{1}{4} \times 2 = 12\frac{1}{2} \text{ units}^2$



Florida's assessment limits for MA.6.GR.1.2 and MA.6.GR.1.3 state that all ordered pairs should be integers. The rational ordered pairs in 4.6.3 and 4.6.5 help to support the area of emphasis, performing all four operations with integers, positive decimals, and positive fractions with procedural fluency, while also continuing student understanding of finding distance and area. By stretching a student's understanding beyond integers, the student will not necessarily be able to rely on counting to determine the distance, thus solidifying his/her understanding of absolute value as the distance from zero.



Consider offering graph paper for students to draw and label the rectangle with its dimensions after the length and width are found. Students may be more hesitant to engage and make mistakes if they only have one coordinate plane to use.



Encourage students to use the strategies they practiced in the previous lessons to determine length and width before finding the area. Draw attention to the scale and ensure students see that the points do not fall directly on the lines. This may be difficult for some students to see because the points are so close to the lines.



For some students, it may be helpful to substitute the coordinates with friendlier integers to allow them to focus on the process of finding distance and area rather than operations with fractions. This will provide an additional entry point without deterring from the learning goals of the lesson.

4.6.4 Collaboration: Finding the Fourth Vertex

Component Overview: Students collaborate to discuss all the attributes they know about rectangles. Prompt students to think about things like side length. Students will apply these ideas to determine where the missing vertex of a rectangle should be placed.



4.6.4 Collaboration: Finding the Fourth Vertex

- With your partner, list every attribute you know about rectangles.

Rectangles

Answers will vary.

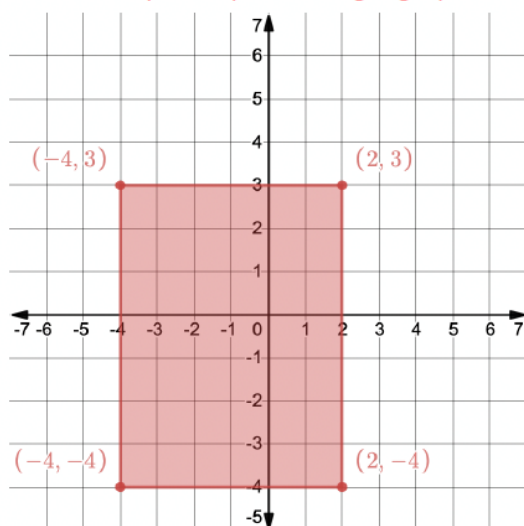
I know that rectangles have two sets of parallel sides.

I know that rectangles have two sets of equal sides.

I know rectangles have four right angles.

- With your partner, find another pair of partners. Share what you wrote about rectangles. Add any new attributes the other group found that you didn't.
- On the coordinate grid, draw a rectangle whose sides lie on the gridlines. Label the ordered pair of each vertex.

Answers will vary. Sample rectangle graphed below.



Highlight student responses that include vocabulary like:

- Parallel
- Equivalent/equal
- Side length
- 90 degrees



Before students begin, consider reviewing the definition of attributes to ensure all students understand the prompt. It may be particularly helpful to review attributes of something students are already familiar with, such as a classroom object, to provide additional examples and nonexamples of attributes.

4.6.4 Collaboration: Finding the Fourth Vertex (continued)

4. Discuss with your partner how to determine where to place the vertices of your rectangle. Write a summary of your discussion.

Answers will vary.

We knew that the horizontal lines both needed to have the same y -coordinates, and the vertical lines both needed to have the same x -coordinates.

5. Based on the rectangle drawn in the previous questions, add any attributes about rectangles to your list.

Answers will vary.



For students who finish early, consider an extension in which students find the area and perimeter of the rectangle they drew.

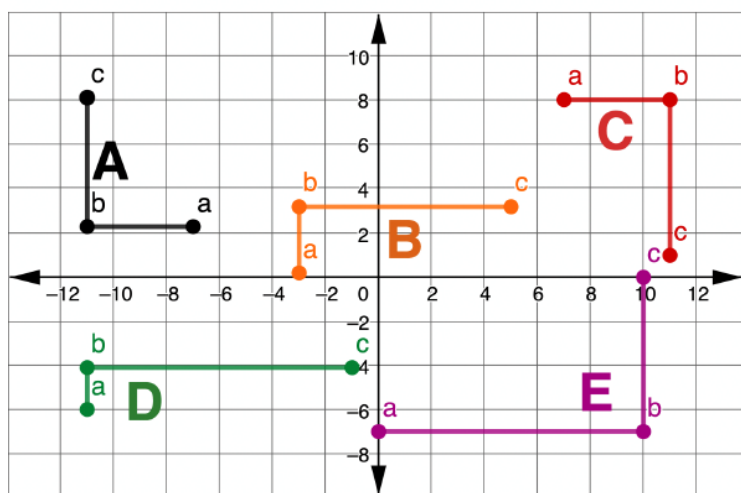
4.6.5 Guided Practice: Finding the Fourth Vertex

Component Overview: Students find the missing vertex in a rectangle on the coordinate plane. Students apply what they know about the attributes of a rectangle as discussed in the 4.6.4 Collaboration to finish each rectangle on the grid.



4.6.5 Guided Practice: Finding the Fourth Vertex

- Using the coordinate plane, find the fourth vertex of the following rectangles.



	Vertex a	Vertex b	Vertex c	Vertex d
Rectangle A	(-7, 2.3)	(-11, 2.3)	(-11, 8.1)	(-7, 8.1)
Rectangle B	(-3, 0.2)	(-3, 3.2)	(5, 3.2)	(5, 0.2)
Rectangle C	(7, 8)	(11, 8)	(11, 1)	(7, 1)
Rectangle D	(-11, -6)	(-11, -4.1)	(-1, -4.1)	(-1, -6)
Rectangle E	(0, -7)	(10, -7)	(10, 0)	(0, 0)

- What is the area of Rectangle A, in square units?

$$\text{Area of rectangle} = 4 \times 5.8 = 23.2 \text{ units}^2$$



Encourage students to apply what they know about the attributes of rectangles to determine the missing points.

- If the sets of lines must be parallel, where can you place the final vertex to create a parallel line?
- How can you use the coordinates of each point to determine the coordinates of the fourth point?
- Where do you need to place the vertex to create a right angle? Why?

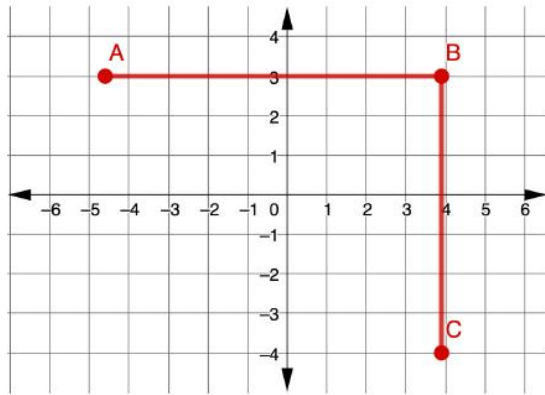
4.6.6 Your Turn!

Component Overview: Students find the missing vertices of the rectangles and then determine the area of each rectangle.



4.6.6 Your Turn!

1. Three vertices of a rectangle are at points $A(-4\frac{3}{5}, 3)$, $B(3\frac{9}{10}, 3)$, and $C(3\frac{9}{10}, -4)$. Each gridline represents 1 centimeter.



- a. What are the coordinates of the fourth vertex?

$$(-4\frac{3}{5}, -4)$$

- b. What is the area of the rectangle?

$$\text{Length of rectangle} = 7 \text{ cm}$$

$$\text{Width of rectangle} = 8\frac{1}{2} \text{ cm}$$

$$\text{Area of rectangle} = 7 \times 8\frac{1}{2} = 59.5 \text{ cm}^2$$

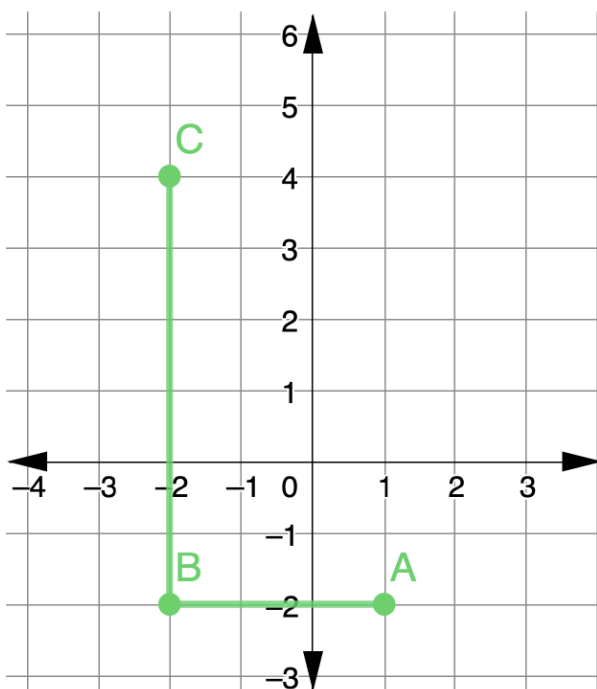


Consider challenging early finishers with the following questions:

- How could you justify that the shape is not a square?
- How would the vertices need to change for the shape to be a square?

4.6.6 Your Turn! (continued)

2. Jada needs to cut out a piece of fabric in the shape of a rectangle. She laid out the fabric on a cutting board, whose scale is represented in feet (ft). Jada placed two edges of the fabric at points A and B . She wants the length of the fabric to be as long as the distance from point B to point C .



- a. Determine the coordinates of the fourth corner of the fabric.
The fourth coordinate would need to be on the point (1,4).
- b. Find the area of the piece of fabric.
Length of fabric = 3 ft
Width of fabric = 6 ft
Area of fabric = $3 \times 6 = 18 \text{ ft}^2$



Encourage students to use the graph and ask themselves “Does my answer make sense?” When applicable, students can count the boxes inside the graph to check their answer for area.

4.6.7 Wrap-Up: Reflection

Component Overview: Use the Wrap-Up to help determine which topics should be reviewed prior to the unit assessment.



4.6.7 Wrap-Up: Reflection

1. Complete the following statements.
 - a. Today, I am still unsure about ...
Answers will vary.
 - b. To fill this gap, I intend to ...
Answers will vary.



Use the 4.6.7 Wrap-Up to help determine which topics may need remediation prior to the unit assessment.